

SAFETY DATA SHEET

According to REACH Regulation (EC) No 1907/2006, as amended by UK REACH Regulations SI 2019/758

CLARIANT 

EMULSOGEN IT

Page 1(58)

Substance key: SXR005008

Revision Date: 23.09.2025

Date of last issue: 14.02.2023

Version : 5 - 2 / GB

Date of printing : 05.04.2026

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Trade name

EMULSOGEN IT

Material number: 107687

Chemical nature:

mixture of calcium alkylbenzenesulphonate, ethoxylated vegetable oil and solubilizer

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses of the substance or mixture

Industry sector :

Crop Protection

Type of use:

Emulsifier for the formulation of plant protection products

Exposure scenarios: see annex

1.3. Details of the supplier of the safety data sheet

Identification of the company

Clariant Produkte (Deutschland) GmbH

65926 Frankfurt am Main

Telephone no. : +49 69 305 18000

Information about the substance/mixture

BU Care Chemicals

Product Stewardship

e-mail: SDS.Europe@clariant.com

1.4. Emergency telephone number

00800-5121 5121 (24 h)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008) as amended by GB-CLP Regulation, UK SI 2019/720, and UK SI 2020/1567)

Flammable liquids, Category 3

H226: Flammable liquid and vapour.

Skin irritation, Category 2

H315: Causes skin irritation.

Serious eye damage, Category 1

H318: Causes serious eye damage.

Long-term (chronic) aquatic hazard, Category 2

H411: Toxic to aquatic life with long lasting effects.

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2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008) as amended by GB-CLP Regulation, UK SI 2019/720, and UK SI 2020/1567)

Hazard pictograms :



Signal word :

Danger

Hazard statements :

H226 Flammable liquid and vapour.
H315 Causes skin irritation.
H318 Causes serious eye damage.
H411 Toxic to aquatic life with long lasting effects.

Precautionary statements :

Prevention:
P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.

P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

P391 Collect spillage.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Hazardous components which must be listed on the label:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts
Iso-Butanol

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Ecological information: This substance/mixture does not contain components considered to have endocrine disrupting properties for environment according to UK REACH Article 57(f) at levels of 0.1% or higher.

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Toxicological information: This substance/mixture does not contain components considered to have endocrine disrupting properties for human health according to UK REACH Article 57(f), at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts	68953-96-8 273-234-6	Acute Tox. 4; H312 Skin Irrit. 2; H315 Eye Dam. 1; H318 Aquatic Chronic 2; H411	>= 30 - < 50
Iso-Butanol	78-83-1 201-148-0 603-108-00-1	Flam. Liq. 3; H226 Skin Irrit. 2; H315 Eye Dam. 1; H318 STOT SE 3; H335 (Respiratory system) STOT SE 3; H336 (Central nervous system)	>= 15 - < 20

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

- General advice : Move out of dangerous area.
Consult a physician.
Show this safety data sheet to the doctor in attendance.
Do not leave the victim unattended.
- If inhaled : Move to fresh air.
If unconscious, place in recovery position and seek medical advice.
Never give anything by mouth to an unconscious person.
Get immediate medical advice/ attention.
- In case of skin contact : If on skin, rinse well with water.
Take off all contaminated clothing immediately.
Take victim immediately to hospital.
- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids,

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for at least 15 minutes.
Get immediate medical advice/ attention.
If easy to do, remove contact lens, if worn.
Protect unharmed eye.
Keep eye wide open while rinsing.

If swallowed : Rinse mouth with water.
Do NOT induce vomiting.
Never give anything by mouth to an unconscious person.
Take victim immediately to hospital.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms : The possible symptoms known are those derived from the
labelling (see section 2).

Risks : The possible health hazards known are those derived from the
labelling (see corresponding section) and/or provided in this
section.

Causes skin irritation.
Causes serious eye damage.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Water spray jet
Alcohol-resistant foam
Dry powder
Carbon dioxide (CO₂)
Use extinguishing measures that are appropriate to local
circumstances and the surrounding environment.

Unsuitable extinguishing media : High volume water jet

5.2 Special hazards arising from the substance or mixture

Hazardous combustion products : Carbon oxides
Hydrocarbons
Sulphur compounds

5.3 Advice for firefighters

Special protective equipment for firefighters : Wear full protective clothing and self-contained breathing
apparatus.

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Further information : In the event of fire and/or explosion do not breathe fumes.
Do not allow run-off from fire fighting to enter drains or water courses.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Avoid contact with skin, eyes and clothing.
Ensure adequate ventilation.
Evacuate personnel to safe areas.
Remove all sources of ignition.
Use personal protective equipment.
Refer to protective measures listed in sections 7 and 8.
Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.

6.2 Environmental precautions

Environmental precautions : The product should not be allowed to enter drains, water courses or the soil.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Prevent product from entering drains.
Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13).
Non-sparking tools should be used.
Take measures to prevent the build up of electrostatic charge.
Clean contaminated surface thoroughly.
Treat recovered material as described in the section "Disposal considerations".

6.4 Reference to other sections

Information regarding Safe handling, see chapter 7., For personal protection see section 8., For disposal considerations see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Use only with adequate ventilation/personal protection.
For personal protection see section 8.
Avoid contact with skin, eyes and clothing.
Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
Keep away from heat, hot surfaces, sparks, open flames and

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other ignition sources. No smoking.

Advice on protection against fire and explosion : Keep away from heat and sources of ignition. Observe the general rules of industrial fire protection Vapours may form explosive mixture with air. Vapours may spread long distances and ignite. Electrical equipment should be protected to the appropriate standard.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Use protective skin cream before handling the product. Wash hands before breaks and at the end of workday. Take off immediately all contaminated clothing and wash it before reuse.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Keep containers tightly closed in a dry, cool and well-ventilated place. Handle and open container with care. Keep away from sources of ignition - No smoking.

Further information on storage conditions : - sensitive to frost - In case of the product becoming opaque, thickening or being frozen due to the effects of cold, allow to thaw slowly at room temperature. Stir briefly before use.

Advice on common storage : When used and handled as intended, none.

Further information on storage stability : Stable under recommended storage conditions.

7.3 Specific end use(s)

Specific use(s) : No further recommendations.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Components	CAS-No.	Value type (Form of exposure)	Control parameters	Basis
Iso-Butanol	78-83-1	STEL	75 ppm 231 mg/m3	GB EH40
		TWA	50 ppm 154 mg/m3	GB EH40

Derived No Effect Level (DNEL):

Substance name	End Use	Exposure routes	Potential health effects	Value
Iso-Butanol CAS-No.: 78-83-1	Workers	Inhalation	Long-term local effects	310 mg/m3
Remarks:DNEL				

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	General population	Inhalation	Long-term local effects	55 mg/m3
	Remarks:DNEL			
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts CAS-No.: 68953-96-8	Workers	Inhalation	Long-term systemic effects	6 mg/m3
	Remarks:DNEL			
	Workers	Dermal	Long-term systemic effects	0.85 mg/kg bw/day
	Remarks:DNEL			
	Consumers	Inhalation	Long-term systemic effects	1.48 mg/m3
	Remarks:DNEL			
	Consumers	Dermal	Long-term systemic effects	0.425 mg/kg bw/day
	Remarks:DNEL			
	Consumers	Oral	Long-term systemic effects	0.43 mg/kg bw/day
	Remarks:DNEL			

Predicted No Effect Concentration (PNEC):

Substance name	Environmental Compartment	Value
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts CAS-No.: 68953-96-8	Fresh water	0.023 mg/l
	Water (intermittent release)	0.29 mg/l
	Marine water	0.002 mg/l
	Sewage treatment plant	5.5 mg/l
	Fresh water sediment	1.35 mg/kg dry weight (d.w.)
	Marine sediment	0.135 mg/kg dry weight (d.w.)
	Soil	0.124 mg/kg dry weight (d.w.)

8.2 Exposure controls

Engineering measures

Use engineering controls such as local or general exhaust to maintain airborne concentrations below exposure limits.

Personal protective equipment

Eye/face protection : Chemical splash goggles with face shield.
Do not wear contact lenses.

Hand protection

Break through time : 480 min
Glove thickness : 0.7 mm
Remarks : Long-term exposure Impervious butyl rubber gloves

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Break through time : 30 min
Glove thickness : 0.4 mm
Remarks : For short-term exposure (splash protection): Nitrile rubber gloves.

Remarks : These types of protective gloves are offered by various manufacturers. Please note the manufacturers' detailed statements, especially about the minimum thickness and the minimum breakthrough time. Consider also the particular working conditions under which the gloves are being used.

Skin and body protection : Wear protective clothing, including long sleeves and gloves, to prevent skin contact.

Respiratory protection : Use respiratory protection unless adequate local exhaust ventilation is provided or exposure assessment demonstrates that exposures are within recommended exposure guidelines. Equipment should conform to EN 14387

Filter type : Combined particulates, inorganic and acidic gas/vapour, ammonia/amines and organic vapour type (ABEK-P)

Protective measures : Observe the usual precautions for handling chemicals. Avoid breathing dust or vapour.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance : Liquid

Colour : yellow-brown

Odour : of isobutanol, characteristic

Odour Threshold : no data available

pH : 6 - 8 (20 °C)
Concentration: 5 %
Method: DIN 53996
Ethanol/Water 1:1

pour point : < -5 °C
Method: ISO 3016

Boiling point : > 118 °C
(1,013 hPa)

Flash point : 39 °C
Method: DIN 51755 (closed cup)

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Evaporation rate	:	Not applicable
Upper explosion limit / upper flammability limit	:	12 %(V) Data relate to solvent
Lower explosion limit / Lower flammability limit	:	1.6 %(V) Data relate to solvent
Vapour pressure	:	10.6 hPa (20 °C) Data relate to solvent
Relative vapour density	:	no data available
Relative density	:	no data available
Density	:	1.04 g/cm ³ (20 °C) Method: DIN 51757
Solubility(ies) Water solubility	:	> 33 g/l soluble (15 °C)
Partition coefficient: n-octanol/water	:	Not applicable
Auto-ignition temperature	:	approx. 300 °C
Decomposition temperature	:	> 300 °C Method: DTA
Viscosity Viscosity, kinematic	:	no data available
Flow time	:	> 60 s at 23 °C Method: ISO 2431
Explosive properties	:	Not explosive Method: Expert judgement
Oxidizing properties	:	The substance or mixture is not classified as oxidizing. Method: Expert judgement

9.2 Other information

Metal corrosion rate	:	< 6.25 mm/a Not corrosive to metals
Particle size	:	Does not apply to liquids.
Self-ignition	:	no data available

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SECTION 10: Stability and reactivity

10.1 Reactivity

No dangerous reaction known under conditions of normal use.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : Uncleaned empty vessels may contain product gases which can form explosive mixtures with air.

10.4 Conditions to avoid

Conditions to avoid : Keep away from heat.
Keep away from open flames, hot surfaces and sources of ignition.

10.5 Incompatible materials

Materials to avoid : Strong oxidizing agents

10.6 Hazardous decomposition products

When handled and stored appropriately, no dangerous decomposition products are known

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Not classified

Product:

Acute dermal toxicity : Acute toxicity estimate: > 2,000 mg/kg
Method: Calculation method

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Acute oral toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 401
GLP: yes
Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : Remarks: not required

Acute dermal toxicity : LD50 (Rat, male and female): 1,000 - 1,600 mg/kg
Method: OECD Test Guideline 402
GLP: yes

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Assessment: The component/mixture is moderately toxic after single contact with skin.

Iso-Butanol:

Acute oral toxicity : LD50 (Rat, male and female): 2,830 - 3,350 mg/kg
Method: OECD Test Guideline 401
GLP: yes
Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : LC50 (Rat, male and female): > 18.18 mg/l
Exposure time: 6 h
Test atmosphere: vapour
Method: Other
GLP: yes
Assessment: The substance or mixture has no acute inhalation toxicity

Acute dermal toxicity : LD50 (Rabbit, male and female): 2,460 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Causes skin irritation.

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Species : Rabbit
Exposure time : 4 h
Assessment : Irritating to skin.
Method : Other
Result : Irritating to skin.
GLP : yes

Iso-Butanol:

Species : Rabbit
Exposure time : 4 h
Method : OECD Test Guideline 404
Result : Irritating to skin.
GLP : yes

Serious eye damage/eye irritation

Causes serious eye damage.

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

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Species	: Rabbit
Exposure time	: 1 h - 14 d
Assessment	: Risk of serious damage to eyes.
Method	: Other
Result	: Risk of serious damage to eyes.
GLP	: yes

Iso-Butanol:

Species	: Rabbit
Exposure time	: 24 h
Assessment	: Risk of serious damage to eyes.
Method	: OECD Test Guideline 405
Result	: Risk of serious damage to eyes.
GLP	: yes

Respiratory or skin sensitisation

Skin sensitisation

Not classified

Respiratory sensitisation

Not classified

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Test Type	: Maximisation Test
Exposure routes	: Dermal
Species	: Guinea pig
Method	: OECD Test Guideline 406
Result	: Not a skin sensitizer.
GLP	: yes

Assessment	: Harmful in contact with skin., Causes skin irritation., Causes serious eye damage.
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Iso-Butanol:

Exposure routes	: Skin contact
Method	: QSAR
Result	: Not a skin sensitizer.

Assessment	: Causes skin irritation., Causes serious eye damage.
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Germ cell mutagenicity

Not classified

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

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Genotoxicity in vitro

: Test Type: Ames test
Test system: Salmonella typhimurium
Concentration: 4 - 10000 µg/plate
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes

Test Type: In vitro gene mutation study in mammalian cells
Test system: Chinese hamster ovary cells
Concentration: 0,6 - 60 µg/ml
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes
Remarks: By analogy with a product of similar composition

Genotoxicity in vivo

: Test Type: Micronucleus test
Species: Mouse (male and female)
Strain: NMRI
Application Route: oral (gavage)
Exposure time: 72 h
Dose: 1122 mg/kg
Method: OECD Test Guideline 474
Result: negative
GLP: yes
Remarks: By analogy with a product of similar composition

Germ cell mutagenicity-
Assessment

: In vitro tests did not show mutagenic effects, In vivo tests did not show mutagenic effects

Iso-Butanol:

Genotoxicity in vitro

: Test Type: Ames test
Test system: Salmonella typhimurium
Concentration: 100- 10000 µg/plate
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: No information available.

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster lung cells
Concentration: 11 - 53 mM
Metabolic activation: without
Method: Other
Result: negative
GLP: No information available.

Test Type: In vitro gene mutation study in mammalian cells
Test system: Chinese hamster lung cells
Concentration: <= 107 mM

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Metabolic activation: with and without metabolic activation

Method: Other

Result: negative

GLP: No information available.

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Strain: NMRI
Cell type: Bone marrow
Application Route: oral (gavage)
Exposure time: single iapplication
Dose: 500 - 1000 - 2000 mg/kg
Method: OECD Test Guideline 474
Result: negative
GLP: yes

Germ cell mutagenicity-
Assessment : In vitro tests did not show mutagenic effects, In vivo tests did
not show mutagenic effects

Carcinogenicity

Not classified

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Carcinogenicity - : No information available.
Assessment

Iso-Butanol:

Carcinogenicity - : No information available.
Assessment

Reproductive toxicity

Not classified

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Effects on fertility : Test Type: Three-generation study
Species: Rat, male and female
Strain: Other
Application Route: oral (feed)
Dose: 14 - 70 - 350 mg/kg bw d
Duration of Single Treatment: 720 d
General Toxicity - Parent: NOAEL: 350 mg/kg body weight
General Toxicity F1: NOAEL: 350 mg/kg body weight
General Toxicity F2: NOAEL: 350 mg/kg body weight
Method: Other
GLP: no
Remarks: By analogy with a product of similar composition

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Effects on foetal development : Test Type: Pre-natal
Species: Mouse, female
Strain: CD1
Application Route: oral (gavage)
Dose: 0,2 - 2,0 - 300 - 600 mg/kg
Duration of Single Treatment: 9 d
General Toxicity Maternal: NOAEL: 2 mg/kg body weight
Teratogenicity: NOAEL: 300 mg/kg body weight
Method: Other
GLP: No information available.
Remarks: By analogy with a product of similar composition

Iso-Butanol:

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Strain: Sprague-Dawley
Application Route: Inhalation
Dose: 500 - 1000 - 2500 ppm
Duration of Single Treatment: 6 h
General Toxicity - Parent: NOAEL: ≥ 7.5 mg/l
General Toxicity F1: NOAEL: ≥ 7.5 mg/l
General Toxicity F2: NOAEL: ≥ 7.5 mg/l
Method: OPPTS 870.3800
GLP: yes

Effects on foetal development : Test Type: Pre-natal
Species: Rat
Strain: wistar
Application Route: Inhalation
Dose: 0,5 - 2,5 - 10 mg/l
Duration of Single Treatment: 6 h
General Toxicity Maternal: NOAEL: 10 mg/l
Teratogenicity: NOAEL: 10 mg/l
Method: OECD Test Guideline 414
GLP: yes

Reproductive toxicity - Assessment : No reproductive toxicity to be expected.
No teratogenic effects to be expected.

STOT - single exposure

Not classified

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

Iso-Butanol:

Exposure routes : Inhalation

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Target Organs	:	Respiratory Tract
Assessment	:	May cause respiratory irritation.
Exposure routes	:	Inhalation
Target Organs	:	Central nervous system
Assessment	:	May cause drowsiness or dizziness.

STOT - repeated exposure

Not classified

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Iso-Butanol:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Species	:	Rat, male and female
NOAEL	:	50 mg/kg bw/day
LOAEL	:	250 mg/kg
Application Route	:	oral (feed)
Exposure time	:	84 d
Number of exposures	:	daily in feed
Dose	:	50 - 250 mg/kg bw d
Control Group	:	yes
Method	:	Other
GLP	:	no
Remarks	:	By analogy with a product of similar composition

Species	:	Rat, male and female
NOAEL	:	85 mg/kg bw/day
LOAEL	:	145 mg/kg
Application Route	:	Drinking water
Exposure time	:	270 d
Number of exposures	:	daily in trinking water
Dose	:	85 - 145 - 430 mg/kg bw d
Control Group	:	yes
Method	:	Other
GLP	:	no
Remarks	:	By analogy with a product of similar composition

Iso-Butanol:

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Species : Rat, male and female
NOAEL : > 1450 mg/kg bw/day
Application Route : Drinking water
Exposure time : 90 d
Number of exposures : continuous
Dose : 1000 - 4000 - 16000 ppm
Control Group : yes
Method : OECD Test Guideline 408
GLP : yes

Species : Rat, male and female
NOAEL : ca. 3 mg/l
Application Route : Inhalation
Exposure time : 102 d
Number of exposures : 6 hours/day, 5 days/week
Dose : 250 - 1000 - 2500 ppm
Control Group : yes
Method : Other
GLP : yes

Application Route : Skin contact
Remarks : not tested.

Aspiration toxicity

Not classified

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

No aspiration toxicity classification

Iso-Butanol:

No aspiration toxicity classification

SECTION 12: Ecological information

12.1 Toxicity

Product:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): 76 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
GLP: yes

Toxicity to microorganisms :
Remarks: no data available

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Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): 31.6 mg/l
End point: mortality
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: OECD Test Guideline 203
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 62 mg/l
End point: mortality
Exposure time: 48 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
GLP: yes

Toxicity to algae/aquatic plants : ErC50 (Pseudokirchneriella subcapitata (green algae)): 29 mg/l
End point: Growth rate
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: no
Remarks: By analogy with a product of similar composition

NOEC (Pseudokirchneriella subcapitata (green algae)): 0.5 mg/l
End point: Growth rate
Exposure time: 96 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: no
Remarks: By analogy with a product of similar composition

Toxicity to microorganisms : EC50 (activated sludge): 550 - 760 mg/l
End point: Bacteria toxicity (respiration inhibition)
Exposure time: 3 h
Test Type: aquatic
Analytical monitoring: yes
Method: OECD Test Guideline 209
GLP: no data available

Toxicity to fish (Chronic toxicity) : NOEC: 0.23 mg/l
End point: Other
Exposure time: 72 d
Species: Oncorhynchus mykiss (rainbow trout)
Test Type: flow-through test

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Analytical monitoring: yes
Method: Other
GLP: yes
Remarks: By analogy with a product of similar composition

Toxicity to soil dwelling
organisms

: Test Type: artificial soil
LC50: > 1,000 mg/kg
Exposure time: 14 d
End point: mortality
Species: Eisenia fetida (earthworms)
Method: OECD Test Guideline 207
GLP:yes
Remarks: By analogy with a product of similar composition

Test Type: artificial soil
NOEC: 250 mg/kg
Exposure time: 14 d
End point: Body weight
Species: Eisenia fetida (earthworms)
Method: OECD Test Guideline 207
GLP:yes
Remarks: By analogy with a product of similar composition

Test Type: artificial soil
EC10: 82 - 100 mg/kg
Exposure time: 21 d
End point: Reproduction
Species: Other
Method: ISO 11267
GLP:no data available
Remarks: By analogy with a product of similar composition

Plant toxicity

: EC50: 167 mg/kg
Exposure time: 21 d
End point: Growth
Species: Sorghum bicolor (sorghum)
Analytical monitoring: no
Method: OECD Test Guideline 208
GLP:yes
Remarks: By analogy with a product of similar composition

NOEC: 100 mg/kg
Exposure time: 21 d
End point: Growth
Species: Sorghum bicolor (sorghum)
Analytical monitoring: no
Method: OECD Test Guideline 208
GLP:yes
Remarks: By analogy with a product of similar composition

EC50: 316 mg/kg

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Exposure time: 21 d
End point: Growth
Species: Phaseolus aureus
Analytical monitoring: no
Method: OECD Test Guideline 208
GLP: yes
Remarks: By analogy with a product of similar composition

NOEC: 100 mg/kg
Exposure time: 21 d
End point: Growth
Species: Phaseolus aureus
Analytical monitoring: no
Method: OECD Test Guideline 208
GLP: yes
Remarks: By analogy with a product of similar composition

Sediment toxicity : Remarks: not tested.

Ecotoxicology Assessment

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

Iso-Butanol:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): 1,430 mg/l
End point: mortality
Exposure time: 96 h
Test Type: flow-through test
Analytical monitoring: yes
Method: Other
GLP: no

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia pulex (Water flea)): 1,100 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: no

Toxicity to algae/aquatic plants : ErC50 (Pseudokirchneriella subcapitata (green algae)): 1,799 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

Toxicity to microorganisms : IC50 (Natural microorganism): 670 mg/l
End point: Growth rate

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Exposure time: 16 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: No information available.

Toxicity to fish (Chronic toxicity) : Remarks: not required

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 20 mg/l
End point: Reproduction rate
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Analytical monitoring: yes
Method: Other
GLP: no
Remarks: The details of the toxic effect relate to the nominal concentration.

12.2 Persistence and degradability

Product:

Biodegradability : Biodegradation: 20 - 70 %
Method: OECD Test Guideline 302B

Chemical Oxygen Demand (COD) : 2,100 mg/g

Dissolved organic carbon (DOC) : 600 mg/g

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Biodegradability : Test Type: aerobic
Inoculum: activated sludge
Concentration: 22 mg/l
Result: Not readily biodegradable.
Biodegradation: 2.9 %
Related to: Carbon dioxide (CO₂)
Exposure time: 28 d
Method: OECD Test Guideline 301E
GLP: no

Physico-chemical removability : Remarks: Not readily biodegradable.

Stability in water : Test Type: abiotic
Degradation half life (Aqueous buffer solution.): > 365 d (50 °C)
pH: 4

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Hydrolysis: (16.7 %)

Method: OECD Test Guideline 111

GLP: yes

Test Type: abiotic

Degradation half life (Aqueous buffer solution.): > 365 d (50 °C)

pH: 7

Hydrolysis: (25.6 %)

Method: OECD Test Guideline 111

GLP: yes

Test Type: abiotic

Degradation half life (Aqueous buffer solution.): > 365 d (50 °C)

pH: 9

Hydrolysis: (18.6 %)

Method: OECD Test Guideline 111

GLP: yes

Iso-Butanol:

Biodegradability

: Test Type: aerobic

Inoculum: activated sludge

Result: Readily biodegradable.

Biodegradation: 70 - 80 %

Related to: BOD in % of theoretical OD

Exposure time: 28 d

Method: OECD Test Guideline 301D

GLP: no

12.3 Bioaccumulative potential

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Bioaccumulation

: Species: Other

Bioconcentration factor (BCF): 3.16

Method: Other

GLP: no

Remarks: Does not bioaccumulate.

The value is given based on a SAR/AAR approach using OECD Toolbox, DEREK, VEGA QSAR models (CAESAR models), etc.

Partition coefficient: n-octanol/water

: log Pow: 4.595 (20 °C)

Method: Regulation (EC) No. 440/2008, Annex, A.8

GLP: yes

Iso-Butanol:

Bioaccumulation

: Remarks: Due to the low logPow bioaccumulation is not

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expected

Partition coefficient: n-
octanol/water : log Pow: 1 (25 °C)
Method: OECD Test Guideline 117
GLP: yes

12.4 Mobility in soil

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Distribution among : adsorption
environmental compartments Medium: Soil
log Koc: > 5
Method: OECD Test Guideline 121
Remarks: Based upon the calculated log Koc, adsorption to
the soil phase is not expected.

Iso-Butanol:

Distribution among : Adsorption/Soil
environmental compartments Medium: water - soil
log Koc: 0.47
Method: estimated

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered
to be either persistent, bioaccumulative and toxic (PBT), or
very persistent and very bioaccumulative (vPvB) at levels of
0.1% or higher.

12.6 Other adverse effects

Product:

Endocrine disrupting : This substance/mixture does not contain components
potential considered to have endocrine disrupting properties for
environment according to UK REACH Article 57(f) at levels of
0.1% or higher.

Components:

Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:

Environmental fate and : not available
pathways

Iso-Butanol:

Environmental fate and : not available
pathways

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Additional ecological information : Do not allow to enter ground water, waterways or waste water.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : Product should be taken to a suitable and authorized waste disposal site in accordance with relevant regulations and if necessary after consultation with the waste disposal operator and/or the competent Authorities

Contaminated packaging : Packaging that cannot be cleaned should be disposed of as product waste

SECTION 14: Transport information

Section 14.1. to 14.5.

ADR

UN no. UN 1993
Proper shipping name: Flammable liquid, n.o.s.
Hazard inducer(s): Isobutanol
Tetrapropylene benzene sulfonate calcium-salt
Class: 3
Primary risk: 3
Packing group: III
Hazard no. : 30
Environmental hazards: Special marking provision: environmentally hazardous
Remarks Shipment permitted

ADN

UN no. UN 1993
Proper shipping name: Flammable liquid, n.o.s.
Hazard inducer(s): Isobutanol
Tetrapropylene benzene sulfonate calcium-salt
Class: 3
Primary risk: 3
Packing group: III
Environmental hazards: Special marking provision: environmentally hazardous
Remarks Shipment permitted

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RID

UN no.	UN 1993
Proper shipping name:	Flammable liquid, n.o.s.
Hazard inducer(s):	Isobutanol
	Tetrapropylene benzene sulfonate calcium-salt
Class:	3
Primary risk:	3
Packing group:	III
Hazard no. :	30
Environmental hazards:	Special marking provision: environmentally hazardous
Remarks	Shipment permitted

IATA

UN no.	UN 1993
Proper shipping name:	Flammable liquid, n.o.s.
Hazard inducer(s):	Isobutanol
	Tetrapropylene benzene sulfonate calcium-salt
Class:	3
Primary risk:	3
Packing group:	III
Environmental hazards:	Special marking provision: environmentally hazardous
Remarks	Shipment permitted

IMDG

UN no.	UN 1993
Proper shipping name:	Flammable liquid, n.o.s.
Hazard inducer(s):	Isobutanol
	Tetrapropylene benzene sulfonate calcium-salt
Class:	3
Primary risk:	3
Packing group:	III
Remarks	Shipment permitted
Marine pollutant:	Marine Pollutant
EmS :	F-E S-E

14.6. Special precautions for user

See sections 6 to 8 of this Safety Data Sheet.

14.7. Maritime transport in bulk according to IMO instruments

No transport as bulk according IBC - Code.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Relevant EU provisions transposed through retained EU law

UK REACH List of restrictions (Annex 17)

: Conditions of restriction for the following entries should be considered:

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UK REACH Candidate list of substances of very high concern (SVHC) for Authorisation	:	Number on list 3 Not applicable
The Persistent Organic Pollutants Regulations (retained Regulation (EU) 2019/1021 as amended for Great Britain)	:	Not applicable
Regulation (EU) No 2024/590 on substances that deplete the ozone layer	:	Not applicable
Council Regulation (EC) No 111/2005 laying down rules for the monitoring of trade between the Community and third countries in drug precursors	:	Neither banned nor restricted
Council Regulation (EC) No 273/2004 on drug precursors	:	Not applicable
UK REACH List of substances subject to authorisation (Annex XIV)	:	Not applicable
GB Export and import of hazardous chemicals - Prior Informed Consent (PIC) Regulation	:	Not applicable
Control of Major Accident Hazards Regulations 2015 (COMAH)	P5c	FLAMMABLE LIQUIDS
	E2	ENVIRONMENTAL HAZARDS

Other regulations:

Take note of The Management of Health and Safety at Work Regulations 1999 (requirements relating to new and expectant mothers at work contained in Regulation 16 to 18) and of the Pregnant Workers Directive 92/85/EEC.

Take note of The Management of Health and Safety at Work Regulations 1999 (requirements relating to protection of young people at work contained in Regulation 19) and of Directive 94/33/EC on the protection of young people at work.

Regulation (EU) No 528/2012 of the European Parliament and of the Council of 22 May 2012 concerning the making available on the market and use of biocidal products

15.2 Chemical safety assessment

Chemical Safety Assessments (CSAs) are available for one or more of the component substances contained in this product.

SECTION 16: Other information

Full text of H-Statements

H226	:	Flammable liquid and vapour.
H312	:	Harmful in contact with skin.

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H315	: Causes skin irritation.
H318	: Causes serious eye damage.
H335	: May cause respiratory irritation.
H336	: May cause drowsiness or dizziness.
H411	: Toxic to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	: Acute toxicity
Aquatic Chronic	: Long-term (chronic) aquatic hazard
Eye Dam.	: Serious eye damage
Flam. Liq.	: Flammable liquids
Skin Irrit.	: Skin irritation
STOT SE	: Specific target organ toxicity - single exposure
GB EH40	: UK. EH40 WEL - Workplace Exposure Limits
GB EH40 / TWA	: Long-term exposure limit (8-hour TWA reference period)
GB EH40 / STEL	: Short-term exposure limit (15-minute reference period)

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

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Further information

Other information : Observe national and local legal requirements
Items where changes have been made to the previous version
are highlighted in the body of this document by two vertical
lines.

Classification of the mixture:

Flam. Liq. 3	H226
Skin Irrit. 2	H315
Eye Dam. 1	H318
Aquatic Chronic 2	H411

Classification procedure:

Based on product data or assessment
Calculation method
Calculation method
Calculation method

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Exposure scenario

Number	Title
ES 1	Formulation or re-packing; Use in agrochemicals PC12, PC27 - PROC2, PROC3, PROC4, PROC5, PROC8b, PROC9, PROC14 - ERC2 Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts
ES 2	Widespread use by professional workers; Use in agrochemicals PC12, PC27 - PROC8a, PROC8b, PROC11 - ERC8a, ERC8d Benzenesulfonic acid, C10-13-alkyl derivs., calcium salt
ES 3	Consumer use; Use in agrochemicals PC12, PC27 - ERC8d Benzenesulfonic acid, C10-13-alkyl derivs., calcium salt

1. ES 1: Formulation or re-packing; Use in agrochemicals

1.1. Title section

Fertilizers (PC12)	
Plant protection products (PC27)	
Environment	
CS1: Formulation or re-packing (Formulation into mixture)	ERC2
Workers	
CS2: Formulation or re-packing (Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions)	PROC2
CS3: Formulation or re-packing (Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions)	PROC2
CS4: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition)	PROC3
CS5: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition)	PROC3
CS6: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition)	PROC3
CS7: Formulation or re-packing (Chemical production where opportunity for exposure arises)	PROC4
CS8: Formulation or re-packing (Chemical production where opportunity for exposure arises)	PROC4
CS9: Formulation or re-packing (Chemical production where opportunity for exposure arises)	PROC4

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CS10: Formulation or re-packing (Mixing or blending in batch processes)	PROC5
CS11: Formulation or re-packing (Mixing or blending in batch processes)	PROC5
CS12: Formulation or re-packing (Transfer of substance or mixture (charging/discharging) at dedicated facilities)	PROC8b
CS13: Formulation or re-packing (Transfer of substance or mixture (charging/discharging) at dedicated facilities)	PROC8b
CS14: Formulation or re-packing (Transfer of substance or mixture into small containers (dedicated filling line, including weighing))	PROC9
CS15: Formulation or re-packing (Tabletting, compression, extrusion, pelettisation, granulation)	PROC14
CS16: Formulation or re-packing (Tabletting, compression, extrusion, pelettisation, granulation)	PROC14
CS17: Formulation or re-packing (Tabletting, compression, extrusion, pelettisation, granulation)	PROC14

1.2. ES 1 Conditions of use affecting exposure

1.2.1 ES 1 - CS 1: Control of environmental exposure: Formulation or re-packing (Formulation into mixture) (ERC2)

Amount used

Daily amount per site : 3 tonnes/day
Annual amount per site : 300 tonnes/year

Environment factors not influenced by risk management

Flow rate of receiving surface : 18,000 m3/d
water

Conditions and measures related to sewage treatment plant

Type of Sewage Treatment Plant : Municipal Sewage Treatment Plant
Flow rate of sewage treatment : 2,000 m3/d
plant effluent
Effectiveness (of a measure) : 9.966 %

Waste management measures

Waste treatment : Contain and dispose of waste according to local regulations.

1.2.2 ES 1 - CS 2: Control of worker exposure: Formulation or re-packing (Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions) (PROC2)

Remarks : Product sampling

Product characteristics

Concentration of the Substance in : <= 90 %
Mixture/Article

Remarks : Liquid, Covers use at ambient temperatures.

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Frequency and duration of use

Exposure duration : 30 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Application rate : < 0.01 L/min, Material transfers

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation. Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure regular inspection, cleaning and maintenance of equipment and machines.
Ensure operatives are trained to minimise exposures.
Regular cleaning of work area
Regular cleaning of equipment
Avoid contact with contaminated tools and objects.
Automate activity where possible.
Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.3 ES 1 - CS 3: Control of worker exposure: Formulation or re-packing (Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions) (PROC2)

Remarks : Equipment cleaning and maintenance

Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 60 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation. Supervision in place to check that the risk management measures in place are being used correctly and operation

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conditions followed.
Ensure control measures are regularly inspected and maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible.
Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.4 ES 1 - CS 4: Control of worker exposure: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition) (PROC3)

Remarks : Charging
Discharging

Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 120 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Application rate : < 1000 L/min, Material transfers

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible.
Technical conditions and measures : Medium-level containment

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Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.5 ES 1 - CS 5: Control of worker exposure: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition) (PROC3)

Remarks : Process sampling

Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 30 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Application rate : < 0.01 L/min, Material transfers

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.

Ensure control measures are regularly inspected and maintained.

Avoid contact with contaminated tools and objects.

Regular cleaning of work area

Regular cleaning of equipment

Ensure operatives are trained to minimise exposures.

Automate activity where possible.

Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.6 ES 1 - CS 6: Control of worker exposure: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition) (PROC3)

Remarks : Equipment cleaning and maintenance

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Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 60 min

Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation. Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.

Ensure control measures are regularly inspected and maintained.

Avoid contact with contaminated tools and objects.

Regular cleaning of work area

Regular cleaning of equipment

Ensure operatives are trained to minimise exposures.

Automate activity where possible.

Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.7 ES 1 - CS 7: Control of worker exposure: Formulation or re-packing (Chemical production where opportunity for exposure arises) (PROC4)

Remarks : Charging
Discharging

Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 120 min

Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Application rate : > 1000 L/min, Material transfers

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Risk Management Measures

Organisational measures to
prevent /limit releases, dispersion
and exposure

: Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management
measures in place are being used correctly and operation
conditions followed.
Ensure control measures are regularly inspected and
maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible.

Technical conditions and
measures

: Semi-closed system

Personal protective measures

: Wear chemically resistant gloves (tested to EN374) in
combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.8 ES 1 - CS 8: Control of worker exposure: Formulation or re-packing (Chemical production where opportunity for exposure arises) (PROC4)

Remarks

: Process sampling

Product characteristics

Concentration of the Substance in
Mixture/Article

: <= 90 %

Remarks

: Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration

: 30 min

Remarks

: ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor

: Indoor use

Application rate

: < 0.1 L/min, Material transfers

Risk Management Measures

Organisational measures to
prevent /limit releases, dispersion
and exposure

: Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management
measures in place are being used correctly and operation
conditions followed.
Ensure control measures are regularly inspected and
maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment

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Personal protective measures : Ensure operatives are trained to minimise exposures.
Automate activity where possible.
: Wear chemically resistant gloves (tested to EN374) in
combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.9 ES 1 - CS 9: Control of worker exposure: Formulation or re-packing (Chemical production where opportunity for exposure arises) (PROC4)

Remarks : Equipment cleaning and maintenance

Product characteristics

Concentration of the Substance in : <= 90 %
Mixture/Article

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 60 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Risk Management Measures

Organisational measures to : Open windows during application to ensure natural ventilation.
prevent /limit releases, dispersion : Supervision in place to check that the risk management
and exposure : measures in place are being used correctly and operation
conditions followed.
Ensure control measures are regularly inspected and
maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible.
Personal protective measures : Wear chemically resistant gloves (tested to EN374) in
combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.10 ES 1 - CS 10: Control of worker exposure: Formulation or re-packing (Mixing or blending in batch processes) (PROC5)

Remarks : Mixing

Product characteristics

Concentration of the Substance in : <= 90 %
Mixture/Article

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Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 420 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Application rate : > 1000 L/min, Material transfers

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation. Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed. Ensure control measures are regularly inspected and maintained. Avoid contact with contaminated tools and objects. Regular cleaning of work area Regular cleaning of equipment Ensure operatives are trained to minimise exposures. Automate activity where possible.

Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training. Wear suitable coveralls to prevent exposure to the skin. Use suitable eye protection.

Effectiveness (of a measure) : 90 %

1.2.11 ES 1 - CS 11: Control of worker exposure: Formulation or re-packing (Mixing or blending in batch processes) (PROC5)

Remarks : Equipment cleaning and maintenance

Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 60 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Risk Management Measures

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- | | | |
|---|---|--|
| Organisational measures to prevent /limit releases, dispersion and exposure | : | Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible. |
| Personal protective measures | : | Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection. |

1.2.12 ES 1 - CS 12: Control of worker exposure: Formulation or re-packing (Transfer of substance or mixture (charging/discharging) at dedicated facilities) (PROC8b)

Remarks : Material transfers

Product characteristics

Concentration of the Substance in Mixture/Article : <= 25 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 420 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Application rate : < 1000 L/min, Material transfers

Risk Management Measures

- | | | |
|---|---|--|
| Organisational measures to prevent /limit releases, dispersion and exposure | : | Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible. |
| Technical conditions and measures | : | Semi-closed system |
| Personal protective measures | : | Wear chemically resistant gloves (tested to EN374) in |

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combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.13 ES 1 - CS 13: Control of worker exposure: Formulation or re-packing (Transfer of substance or mixture (charging/discharging) at dedicated facilities) (PROC8b)

Remarks : Equipment cleaning and maintenance

Product characteristics

Concentration of the Substance in Mixture/Article : <= 25 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 60 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible.

Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.14 ES 1 - CS 14: Control of worker exposure: Formulation or re-packing (Transfer of substance or mixture into small containers (dedicated filling line, including weighing)) (PROC9)

Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

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Exposure duration : 480 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Application rate : 100 - 1000 L/min, Material transfers

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible.

Technical conditions and measures : Semi-closed system

Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.15 ES 1 - CS 15: Control of worker exposure: Formulation or re-packing (Tabletting, compression, extrusion, pelettisation, granulation) (PROC14)

Remarks : Mixing

Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 180 min
Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Surface area : 1 - 3 m²

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion : Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management

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and exposure

measures in place are being used correctly and operation conditions followed.

Ensure control measures are regularly inspected and maintained.

Avoid contact with contaminated tools and objects.

Regular cleaning of work area

Regular cleaning of equipment

Ensure operatives are trained to minimise exposures.

Automate activity where possible.

Personal protective measures

: Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

Wear suitable coveralls to prevent exposure to the skin.

Use suitable eye protection.

1.2.16 ES 1 - CS 16: Control of worker exposure: Formulation or re-packing (Tabletting, compression, extrusion, pelettisation, granulation) (PROC14)

Remarks

: Tabletting, compression, extrusion, pelettisation, granulation

Product characteristics

Concentration of the Substance in Mixture/Article : <= 10 %

Remarks

: Solid, low dustiness, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 240 min

Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Tabletting, compression, extrusion or pelletisation : 100 - 1000 kg/min

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.

Ensure control measures are regularly inspected and maintained.

Avoid contact with contaminated tools and objects.

Regular cleaning of work area

Regular cleaning of equipment

Ensure operatives are trained to minimise exposures.

Automate activity where possible.

Technical conditions and measures

: Semi-closed system

Personal protective measures

: Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.

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Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.2.17 ES 1 - CS 17: Control of worker exposure: Formulation or re-packing (Tabletting, compression, extrusion, pelettisation, granulation) (PROC14)

Remarks : Equipment cleaning and maintenance

Product characteristics

Concentration of the Substance in Mixture/Article : <= 90 %

Remarks : Liquid, Covers use at ambient temperatures.

Frequency and duration of use

Exposure duration : 60 min

Remarks : ART v1.5

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor use

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.

Ensure control measures are regularly inspected and maintained.

Avoid contact with contaminated tools and objects.

Regular cleaning of work area

Regular cleaning of equipment

Ensure operatives are trained to minimise exposures.

Automate activity where possible.

Personal protective measures : Wear chemically resistant gloves (tested to EN374) in combination with 'basic' employee training.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

1.3. ES 1 Exposure estimation and reference to its source

1.3.1 ES 1 - CS 1: Environmental release and exposure: Formulation or re-packing (Formulation into mixture) (ERC2)

Release route	Release rate	Release estimation method
Water	0.18 kg/day	
Air	75 kg/day	
Soil	0.01 %	

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Waste	0 %	
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protection target	Exposure estimation and reference to its source (EUSES v2.1)	RCR
Freshwater	0.008 mg/L	0.36
Freshwater sediment	0.032 mg/kg dry weight	0.02
Marine water	0.000822 mg/L	0.36
Marine sediment	0.003 mg/kg dry weight	0.02
Sewage treatment plant	0.081 mg/L	0.02
Soil	0.001 mg/kg dry weight	0.01
Man via environment - Inhalation	0.006 mg/m ³	< 0.01
Man via environment - Oral	0.026 mg/kg bw/day	0.06
Indirect exposure to humans via the environment		0.07

1.3.2 ES 1 - CS 2: Worker exposure: Formulation or re-packing (Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions) (PROC2)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	0.013 mg/m ³ (ART v1.5, long-term, systemic)	< 0.01
dermal	0.137 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	< 0.01

1.3.3 ES 1 - CS 3: Worker exposure: Formulation or re-packing (Chemical production or refinery in closed continuous process with occasional controlled exposure or processes with equivalent containment conditions) (PROC2)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	0.013 mg/m ³ (ART v1.5, long-term, systemic)	< 0.01
dermal	0.137 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	< 0.01

1.3.4 ES 1 - CS 4: Worker exposure: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition) (PROC3)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	0.078 mg/m ³ (ART v1.5, long-term, systemic)	0.01

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dermal	0.069 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	0.01

1.3.5 ES 1 - CS 5: Worker exposure: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition) (PROC3)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	0.078 mg/m ³ (ART v1.5, long-term, systemic)	0.01
dermal	0.069 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	0.01

1.3.6 ES 1 - CS 6: Worker exposure: Formulation or re-packing (Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition) (PROC3)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	0.078 mg/m ³ (ART v1.5, long-term, systemic)	0.01
dermal	0.069 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	0.01

1.3.7 ES 1 - CS 7: Worker exposure: Formulation or re-packing (Chemical production where opportunity for exposure arises) (PROC4)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	1.9 mg/m ³ (ART v1.5, long-term, systemic)	0.32
dermal	0.686 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	0.33

1.3.8 ES 1 - CS 8: Worker exposure: Formulation or re-packing (Chemical production where opportunity for exposure arises) (PROC4)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	1.9 mg/m ³ (ART v1.5, long-term, systemic)	0.32
dermal	0.686 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01

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	term, systemic)	
combined routes	long-term, systemic	0.33

1.3.9 ES 1 - CS 9: Worker exposure: Formulation or re-packing (Chemical production where opportunity for exposure arises) (PROC4)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	1.9 mg/m ³ (ART v1.5, long-term, systemic)	0.32
dermal	0.686 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	0.33

1.3.10 ES 1 - CS 10: Worker exposure: Formulation or re-packing (Mixing or blending in batch processes) (PROC5)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	1.1 mg/m ³ (ART v1.5, long-term, systemic)	0.18
dermal	1.371 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	0.02
combined routes	long-term, systemic	0.2

1.3.11 ES 1 - CS 11: Worker exposure: Formulation or re-packing (Mixing or blending in batch processes) (PROC5)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	1.1 mg/m ³ (ART v1.5, long-term, systemic)	0.18
dermal	1.371 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	0.02
combined routes	long-term, systemic	0.2

1.3.12 ES 1 - CS 12: Worker exposure: Formulation or re-packing (Transfer of substance or mixture (charging/discharging) at dedicated facilities) (PROC8b)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	3.4 mg/m ³ (ART v1.5, long-term, systemic)	0.57
dermal	0.823 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	0.58

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1.3.13 ES 1 - CS 13: Worker exposure: Formulation or re-packing (Transfer of substance or mixture (charging/discharging) at dedicated facilities) (PROC8b)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	3.4 mg/m ³ (ART v1.5, long-term, systemic)	0.57
dermal	0.823 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	< 0.01
combined routes	long-term, systemic	0.58

1.3.14 ES 1 - CS 14: Worker exposure: Formulation or re-packing (Transfer of substance or mixture into small containers (dedicated filling line, including weighing)) (PROC9)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	1.2 mg/m ³ (ART v1.5, long-term, systemic)	0.2
dermal	0.686 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	0.081
combined routes	long-term, systemic	0.28

1.3.15 ES 1 - CS 15: Worker exposure: Formulation or re-packing (Tabletting, compression, extrusion, pelettisation, granulation) (PROC14)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	2.1 mg/m ³ (ART v1.5, long-term, systemic)	0.35
dermal	0.343 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	0.04
combined routes	long-term, systemic	0.39

1.3.16 ES 1 - CS 16: Worker exposure: Formulation or re-packing (Tabletting, compression, extrusion, pelettisation, granulation) (PROC14)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	2.1 mg/m ³ (ART v1.5, long-term, systemic)	0.35
dermal	0.343 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	0.04
combined routes	long-term, systemic	0.39

1.3.17 ES 1 - CS 17: Worker exposure: Formulation or re-packing (Tabletting,

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compression, extrusion, pelettisation, granulation) (PROC14)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	2.1 mg/m ³ (ART v1.5, long-term, systemic)	0.35
dermal	0.343 mg/kg bw/day (ECETOC TRA worker v3, long-term, systemic)	0.04
combined routes	long-term, systemic	0.39

1.4. ES 1 Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

2. ES 2: Widespread use by professional workers; Use in agrochemicals

2.1. Title section

Fertilizers (PC12)	
Plant protection products (PC27)	
Environment	
CS1: Widespread use by professional workers (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor))	ERC8a, ERC8d
CS2: Widespread use by professional workers (Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor))	ERC8d
Workers	
CS3: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities, Non-industrial spraying)	PROC8a, PROC11
CS4: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities, Non-industrial spraying)	PROC8a, PROC11
CS5: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities)	PROC8a
CS6: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at dedicated facilities)	PROC8b
CS7: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities)	PROC8a

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2.2. ES 2 Conditions of use affecting exposure

2.2.1 ES 2 - CS 1: Control of environmental exposure: Widespread use by professional workers (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor)) (ERC8a, ERC8d)

Remarks : Granule application / Seed treatment

Conditions and measures related to sewage treatment plant

Type of Sewage Treatment Plant : Municipal Sewage Treatment Plant
Effectiveness (of a measure) : 100 %

Waste management measures

Waste treatment : Contain and dispose of waste according to local regulations.

2.2.2 ES 2 - CS 2: Control of environmental exposure: Widespread use by professional workers (Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor)) (ERC8d)

Remarks : Spraying

Conditions and measures related to sewage treatment plant

Type of Sewage Treatment Plant : Municipal Sewage Treatment Plant
Effectiveness (of a measure) : 91.03 %

Waste management measures

Waste treatment : Contain and dispose of waste according to local regulations.

2.2.3 ES 2 - CS 3: Control of worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities, Non-industrial spraying) (PROC8a, PROC11)

Remarks : Use as a co-formulant in plant protection products, spray application by professionals
Mixing and loading of plant protection products into delivery equipment

Product characteristics

Concentration of the Substance in Mixture/Article : <= 100 %

Physical Form (at time of use) : Liquid
Physical Form (at time of use) : granules, powder

Amount used

Amounts used : 20 kg/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor or outdoor use

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Risk Management Measures

- | | | |
|---|---|--|
| Organisational measures to prevent /limit releases, dispersion and exposure | : | Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible. |
| Technical conditions and measures | : | Provide a basic standard of general ventilation (1 to 3 air changes per hour). |
| Personal protective measures | : | Wear suitable gloves tested to EN374.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection. |

2.2.4 ES 2 - CS 4: Control of worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities, Non-industrial spraying) (PROC8a, PROC11)

- | | | |
|---------|---|--|
| Remarks | : | Use as a co-formulant in plant protection products, spray application by professionals
Delivery and dispersion of plant protection products |
|---------|---|--|

Product characteristics

- | | | |
|---|---|------------------|
| Concentration of the Substance in Mixture/Article | : | <= 100 % |
| Physical Form (at time of use) | : | Liquid |
| Physical Form (at time of use) | : | granules, powder |

Amount used

- | | | |
|--------------|---|----------|
| Amounts used | : | 1 kg/day |
|--------------|---|----------|

Other operational conditions affecting workers exposure

- | | | |
|------------------|---|-----------------------|
| Outdoor / Indoor | : | Indoor or outdoor use |
|------------------|---|-----------------------|

Risk Management Measures

- | | | |
|---|---|---|
| Organisational measures to prevent /limit releases, dispersion and exposure | : | Observe the usage / storage instructions.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible. |
|---|---|---|

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- | | | |
|-----------------------------------|---|---|
| Technical conditions and measures | : | Provide a basic standard of general ventilation (1 to 3 air changes per hour). |
| Personal protective measures | : | Wear suitable gloves tested to EN374.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection. |

2.2.5 ES 2 - CS 5: Control of worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities) (PROC8a)

- | | | |
|---------|---|--|
| Remarks | : | Use as a co-formulant in plant protection products, seed and granular applications by professionals
Mixing and loading of plant protection products into delivery equipment |
|---------|---|--|

Product characteristics

- | | | |
|---|---|----------|
| Concentration of the Substance in Mixture/Article | : | <= 100 % |
|---|---|----------|

- | | | |
|--------------------------------|---|------------------|
| Physical Form (at time of use) | : | Liquid |
| Physical Form (at time of use) | : | granules, powder |

Amount used

- | | | |
|--------------|---|----------------|
| Amounts used | : | 20 kg |
| Remarks | : | granules |
| Amounts used | : | 10 kg |
| Remarks | : | Seed treatment |

Frequency and duration of use

- | | | |
|-------------------|---|-----|
| Exposure duration | : | 8 h |
|-------------------|---|-----|

Other operational conditions affecting workers exposure

- | | | |
|------------------|---|-----------------------|
| Outdoor / Indoor | : | Indoor or outdoor use |
|------------------|---|-----------------------|

Risk Management Measures

- | | | |
|---|---|---|
| Organisational measures to prevent /limit releases, dispersion and exposure | : | Observe the usage / storage instructions.
Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.
Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible. |
| Personal protective measures | : | Wear suitable gloves tested to EN374.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection. |

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2.2.6 ES 2 - CS 6: Control of worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at dedicated facilities) (PROC8b)

Remarks : Use as a co-formulant in plant protection products, seed and granular applications by professionals
Transfer of treated seeds from batch treater into bags

Product characteristics

Concentration of the Substance in Mixture/Article : <= 50 %

Physical Form (at time of use) : powder

Amount used

Amounts used : 10 kg

Frequency and duration of use

Exposure duration : 8 h

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor or outdoor use

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Observe the usage / storage instructions.
Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.

Avoid contact with contaminated tools and objects.
Regular cleaning of work area
Regular cleaning of equipment
Ensure operatives are trained to minimise exposures.
Automate activity where possible.

Technical conditions and measures : Semi-closed system

Personal protective measures : Wear suitable gloves tested to EN374.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

Technical conditions and measures : Local exhaust ventilation

Effectiveness (of a measure) : 90 %

2.2.7 ES 2 - CS 7: Control of worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities) (PROC8a)

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Remarks : Use as a co-formulant in plant protection products, seed and granular applications by professionals
Delivery and dispersion of plant protection products

Concentration of the Substance in Mixture/Article : <= 100 %

Amount used

Amounts used : 20 kg/day

Frequency and duration of use

Exposure duration : 8 h

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor or outdoor use

Risk Management Measures

Organisational measures to prevent /limit releases, dispersion and exposure : Open windows during application to ensure natural ventilation.
Supervision in place to check that the risk management measures in place are being used correctly and operation conditions followed.
Ensure control measures are regularly inspected and maintained.

Avoid contact with contaminated tools and objects.

Regular cleaning of work area

Regular cleaning of equipment

Ensure operatives are trained to minimise exposures.

Automate activity where possible.

Personal protective measures : Wear suitable gloves tested to EN374.
Wear suitable coveralls to prevent exposure to the skin.
Use suitable eye protection.

2.3. ES 2 Exposure estimation and reference to its source

2.3.1 ES 2 - CS 1: Environmental release and exposure: Widespread use by professional workers (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor)) (ERC8a, ERC8d)

Release route	Release rate	Release estimation method
Water	0 %	ECPA SPERC 8d.1.v1
Air	0 %	ECPA SPERC 8d.1.v1
Soil	100 %	ECPA SPERC 8d.1.v1
Waste	0.01 %	OECD

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protection target	Exposure estimation and reference to its source (EUSES v2.1)	RCR
Freshwater	0.000128 mg/L	< 0.01
Freshwater sediment	0.0004978 mg/kg dry weight	< 0.01
Marine water	0.0000117 mg/L	< 0.01
Marine sediment	0.0000454 mg/kg dry weight	< 0.01
Sewage treatment plant	0 mg/L	< 0.01
Soil	0.0000007 mg/kg dry weight	< 0.01
Man via environment - Inhalation	0.0000062 mg/m ³	< 0.01
Man via environment - Oral	0.0000239 mg/kg bw/day	< 0.01
Indirect exposure to humans via the environment		< 0.01

2.3.2 ES 2 - CS 2: Environmental release and exposure: Widespread use by professional workers (Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor)) (ERC8d)

Release route	Release rate	Release estimation method
Water	0 %	ECPA SPERC 8d.2.v2
Air	100 %	ECPA SPERC 8d.2.v2
Soil	0 %	ECPA SPERC 8d.2.v2
Waste	0.01 %	ECPA

protection target	Exposure estimation and reference to its source (EUSES)	RCR
Freshwater	0.0000031 mg/L	< 0.01
Freshwater sediment	0.000116 mg/kg dry weight	< 0.01
Marine water	0.0000003 mg/L	< 0.01
Marine sediment	0.0000101 mg/kg dry weight	< 0.01
Sewage treatment plant	0 mg/L	< 0.01
Soil	0.0000008 mg/kg dry weight	< 0.01
Man via environment - Inhalation	0.0000057 mg/m ³	< 0.01
Man via environment - Oral	0.0000005 mg/kg bw/day	< 0.01
Indirect exposure to humans via the environment		< 0.01

2.3.3 ES 2 - CS 3: Worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities, Non-industrial spraying) (PROC8a, PROC11)

Route of exposure and type of effects	Exposure estimate	RCR
dermal	0.070 mg/kg bw/day (ECPA OWB, Automated task)	0.01
inhalative	1.321 mg/m ³ (ECPA OWB, Automated task)	0.16
combined routes	ECPA OWB, Automated task	0.17

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2.3.4 ES 2 - CS 4: Worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities, Non-industrial spraying) (PROC8a, PROC11)

Route of exposure and type of effects	Exposure estimate	RCR
dermal	3.508 mg/kg bw/day (ECPA OWB, Manual)	0.59
inhalative	0.475 mg/m ³ (ECPA OWB, Manual)	0.05
combined routes	ECPA OWB, Manual	0.64

2.3.5 ES 2 - CS 5: Worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities) (PROC8a)

Route of exposure and type of effects	Exposure estimate	RCR
dermal	0.1741 mg/kg bw/day (ECPA OWB)	
inhalative	0.3002 mg/m ³ (ECPA OWB)	
combined routes	0.217 mg/kg bw/day (ECPA OWB)	

2.3.6 ES 2 - CS 6: Worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at dedicated facilities) (PROC8b)

Route of exposure and type of effects	Exposure estimate	RCR
dermal	0.0069 mg/kg bw/day (ECETOC TRA)	
inhalative	1.25 mg/m ³ (ECETOC TRA)	
combined routes	0.1854 mg/kg bw/day (ECETOC TRA)	

2.3.7 ES 2 - CS 7: Worker exposure: Widespread use by professional workers (Transfer of substance or mixture (charging/discharging) at non dedicated-facilities) (PROC8a)

Route of exposure and type of effects	Exposure estimate	RCR
dermal	5.866 mg/kg bw/day (ECPA OWB)	0.07
inhalative	0.274 mg/m ³ (ECPA OWB)	0.046
combined routes	5.905 mg/kg bw/day (ECPA OWB)	0.115

2.4. ES 2 Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

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3. ES 3: Consumer use; Use in agrochemicals

3.1. Title section

Fertilizers (PC12)	
Plant protection products (PC27)	
Environment	
CS1: Consumer use (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor))	ERC8a, ERC8d
CS2: Consumer use (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor))	ERC8a, ERC8d
Consumer	
CS3: Consumer use (Fertilizers)	PC12
CS4: Consumer use (Plant protection products)	PC27

3.2. ES 3 Conditions of use affecting exposure

3.2.1 ES 3 - CS 1: Control of environmental exposure: Consumer use (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor)) (ERC8a, ERC8d)

Remarks : Granule application / Seed treatment

Conditions and measures related to sewage treatment plant

Type of Sewage Treatment Plant : Municipal Sewage Treatment Plant
Effectiveness (of a measure) : 100 %

Waste management measures

Waste treatment : Contain and dispose of waste according to local regulations.

3.2.2 ES 3 - CS 2: Control of environmental exposure: Consumer use (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor)) (ERC8a, ERC8d)

Remarks : Spraying

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Conditions and measures related to sewage treatment plant

Type of Sewage Treatment Plant : Municipal Sewage Treatment Plant
Effectiveness (of a measure) : 100 %

Waste management measures

Waste treatment : Contain and dispose of waste according to local regulations.

3.2.3 ES 3 - CS 3: Control of consumer exposure: Consumer use (Fertilizers) (PC12)

Concentration of the Substance in : <= 0.1 %
Mixture/Article

Human factors not influenced by risk management

Assumes that potential dermal :
contact is limited to hands.

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Observe the usage / storage instructions.
Avoid direct eye contact with product, also via contamination
on hands.

3.2.4 ES 3 - CS 4: Control of consumer exposure: Consumer use (Plant protection products) (PC27)

Product characteristics

Concentration of the Substance in : <= 100 %
Mixture/Article

Physical Form (at time of use) : Liquid
Physical Form (at time of use) : granules, powder

Amount used

Amounts used : 1 kg/ha
Amounts used : 0.02 kg/day

Frequency and duration of use

Frequency of use : 1 uses per day

Other given operational conditions affecting consumers exposure

Outdoor / Indoor : Indoor or outdoor use

Conditions and measures related to protection of consumer (e.g. behavioural advice, personal protection and hygiene)

Consumer Measures : Observe the usage / storage instructions.
Avoid direct eye contact with product, also via contamination
on hands.
Open windows during application to ensure natural ventilation.

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3.3. ES 3 Exposure estimation and reference to its source

3.3.1 ES 3 - CS 1: Environmental release and exposure: Consumer use (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor)) (ERC8a, ERC8d)

Release route	Release rate	Release estimation method
Water	0 %	ECPA SPERC 8d.1.v1
Air	0 %	ECPA SPERC 8d.1.v1
Soil	100 %	ECPA SPERC 8d.1.v1
Waste	0.01 %	OECD

protection target	Exposure estimation and reference to its source (EUSES v2.1)	RCR
Freshwater	0.000129 mg/L	< 0.01
Freshwater sediment	0.000498 mg/kg dry weight	< 0.01
Marine water	0.000012 mg/L	< 0.01
Marine sediment	0.000045 mg/kg dry weight	< 0.01
Sewage treatment plant	0 mg/L	< 0.01
Soil	0.0000007 mg/kg dry weight	< 0.01
Man via environment - Inhalation	0.000006 mg/m ³	< 0.01
Man via environment - Oral	0.000024 mg/kg bw/day	< 0.01
Indirect exposure to humans via the environment		< 0.01

3.3.2 ES 3 - CS 2: Environmental release and exposure: Consumer use (Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor), Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor)) (ERC8a, ERC8d)

Release route	Release rate	Release estimation method
Water	0 %	ECPA SPERC 8d.2.v2
Air	100 %	ECPA SPERC 8d.2.v2
Soil	0 %	ECPA SPERC 8d.2.v2
Waste	0.01 %	ECPA

protection target	Exposure estimation and reference to its source (EUSES v2.1)	RCR
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Substance key: SXR005008

Revision Date: 23.09.2025

Date of last issue: 14.02.2023

Version : 5 - 2 / GB

Date of printing : 05.04.2026

Freshwater	0.000129 mg/L	< 0.01
Freshwater sediment	0.000498 mg/kg dry weight	< 0.01
Marine water	0.000012 mg/L	< 0.01
Marine sediment	0.000045 mg/kg dry weight	< 0.01
Sewage treatment plant	0 mg/L	< 0.01
Soil	0.0000007 mg/kg dry weight	< 0.01
Man via environment - Inhalation	0.000006 mg/m ³	< 0.01
Man via environment - Oral	0.000024 mg/kg bw/day	< 0.01
Indirect exposure to humans via the environment		< 0.01

3.3.3 ES 3 - CS 3: Consumer exposure: Consumer use (Fertilizers) (PC12)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	0 mg/m ³ (ECETOC TRA consumer v3, long-term, systemic)	< 0.01
dermal	0.143 mg/kg bw/day (ECETOC TRA consumer v3, long-term, systemic)	< 0.01
oral	0.017 mg/kg bw/day (ECETOC TRA consumer v3)	0.04
combined routes	ECETOC TRA consumer v3	0.04

3.3.4 ES 3 - CS 4: Consumer exposure: Consumer use (Plant protection products) (PC27)

Route of exposure and type of effects	Exposure estimate	RCR
inhalative	0.02 mg/m ³ (ECETOC TRA consumer v3, long-term, systemic)	< 0.01
dermal	0.206 mg/kg bw/day (ECETOC TRA consumer v3, long-term, systemic)	< 0.01
combined routes	ECETOC TRA consumer v3	< 0.01

3.4. ES 3 Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario